here are open source tools by HashiCorp, Mesos, Docker, Puppet, Chef, Jenkins and many, many more. It's interesting that most of the tooling layer is made up of open source software.

- The management layer is at the top and has the goal of abstracting away both the tooling and infrastructure layers into one common workflow for turning application code into a running application. An example of an end goal would be to just package your application — potentially as a Docker container, *rkt* container, jar or binary — and enable to run on any cloud or physical infrastructure. Ideally, no knowledge of the infrastructure itself is necessary and all the logic is held at the application level.

The infrastructure layer is facing waves of competition and commoditisation, and we now see companies in this space trying to move up the stack into tooling and management. Amazon Web Services (AWS) has released tools like EC2 Container Service (ECS) and AWS CodeDeploy, which allow users to simply package their applications and ignore the infrastructure configuration and provisioning processes. Google Cloud Platform has similarly released Google App Engine and Google Container Service, and sponsors open source projects such as Kubernetes. VMware has moved into tooling and management with heavy support for Pivotal's Cloud Foundry. These tools make it easier for users to deploy applications, and allow infrastructure providers to lock users into their platforms.

The tooling layer is more difficult to monetise, as the tools themselves solve pieces of the greater DevOps goal. It's difficult to charge at enterprise levels for tools that package applications or provision infrastructure. Enterprises pay for tools that reduce costs by saving on employee time or on material. These tools still require significant employee effort and are difficult to tie to material savings; thus, they are harder to monetise. It is certainly possible to monetise this layer through support and services; however, these are more difficult to scale.

The management layer presents the most significant monetisation opportunity as it presents a full solution for turning application code into a running application. The employee investment is significant in the implementation phase, but relatively minimal for upkeep. If the management level can get into material cost savings, such as reducing the size of server fleets, there will be tremendous opportunities for revenue.

DevOps companies are racing to the management layer; however, several tools are necessary to have a complete DevOps management offering.

The three types of DevOps companies

Infrastructure providers, configuration management firms and immutable infrastructure companies are the three types of DevOps organisations.

Figure 1: DevOps workflow

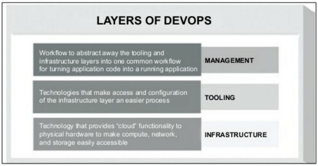

Figure 2: Layers of DevOps

Given that the management layer has the greatest revenue potential, the active players in the space are all battling to capture the opportunity. The interesting aspect of this battle is that the different types of companies in DevOps are approaching it from unique vantage points. These organisations can be categorised as either infrastructure providers, configuration management companies, or companies that favour immutable infrastructure. Amazon Web Services, Google Cloud Platform, OpenStack and VMware are the major players in the infrastructure provider space. Ansible, Chef, Puppet and Salt are the leaders in configuration management. HashiCorp, CoreOS, Docker and Mesosphere are the leaders in the nascent immutable infrastructure space.

All of these companies must have solutions for infrastructure provisioning and application packaging or configuration in order to have a viable chance of winning the battle at the management layer. Historically, VMware is a great example of owning the entire tooling stack to win the datacentre virtualisation market. If you look at the current product development across the companies, you can see that each is building tooling around these essentials. Chef has Chef Provisioning to provision infrastructure, and Chef Configuration Management to configure the application. HashiCorp has Terraform to provision infrastructure and Packer to package applications and build machine images. Docker has Docker Machine to provision infrastructure and Docker to package applications.

These tools are then combined into a dashboard or common workflow to provide a management layer. For example, Mesosphere bundles Mesos, Marathon and Chronos to create the Mesosphere Datacenter Operating System. HashiCorp unites Packer, Terraform and Consul to